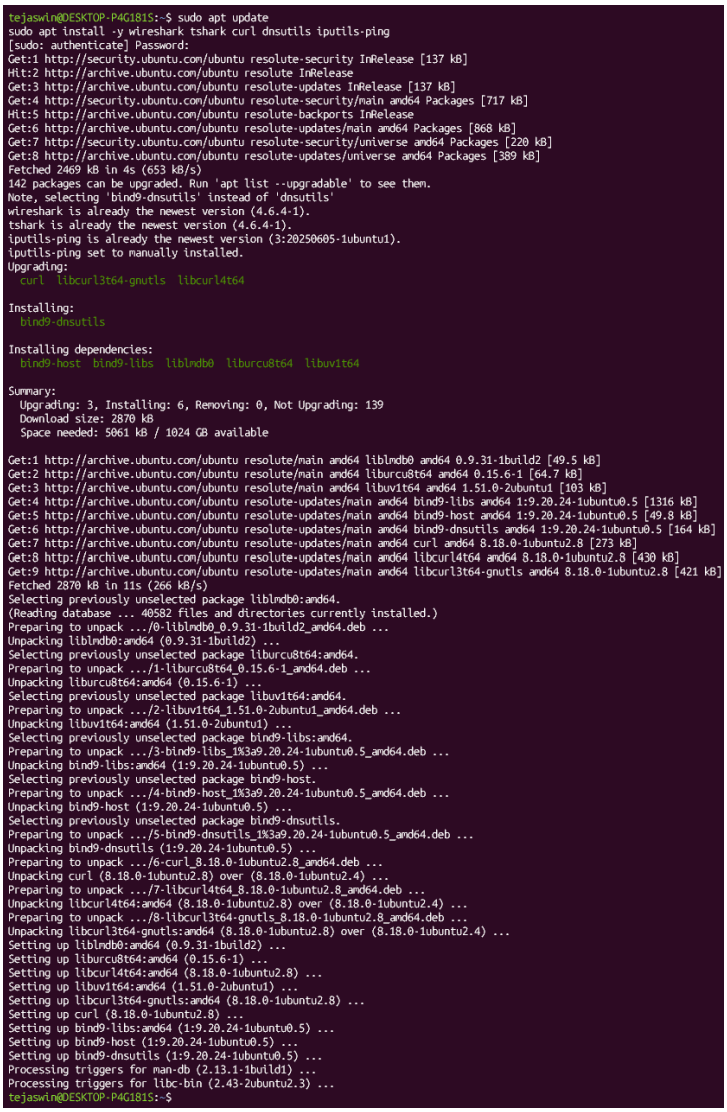
CNNP Skill 1

Name: Tejaswin Amara

Roll No.: 2520090104



```
tejaswin@DESKTOP-P4G181S:~$ sudo apt update
sudo apt install -y wireshark tshark curl dnsutils iputils-ping
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [717 kB]
Hit:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [868 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [220 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [389 kB]
Fetched 2469 kB in 4s (653 kB/s)
142 packages can be upgraded. Run 'apt list --upgradable' to see them.
Note, selecting 'bind9-dnsutils' instead of 'dnsutils'
wireshark is already the newest version (4.6.4-1).
tshark is already the newest version (4.6.4-1).
iputils-ping is already the newest version (3:20250605-1ubuntu1).
iputils-ping set to manually installed.
Upgrading:
  curl  libcurl3t64-gnutls  libcurl4t64

Installing:
  bind9-dnsutils

Installing dependencies:
  bind9-host  bind9-libs  liblmdb0  liburcu8t64  libuv1t64

Summary:
  Upgrading: 3, Installing: 6, Removing: 0, Not Upgrading: 139
  Download size: 2870 kB
  Space needed: 5061 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 liblmdb0 amd64 0.9.31-1build2 [49.5 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 liburcu8t64 amd64 0.15.6-1 [64.7 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libuv1t64 amd64 1.51.0-2ubuntu1 [103 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 bind9-libs amd64 1:9.20.24-1ubuntu0.5 [1316 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 bind9-host amd64 1:9.20.24-1ubuntu0.5 [49.8 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 bind9-dnsutils amd64 1:9.20.24-1ubuntu0.5 [164 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 curl amd64 8.18.0-1ubuntu2.8 [273 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libcurl4t64 amd64 8.18.0-1ubuntu2.8 [430 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libcurl3t64-gnutls amd64 8.18.0-1ubuntu2.8 [421 kB]
Fetched 2870 kB in 11s (266 kB/s)
Selecting previously unselected package liblmdb0:amd64.
(Reading database ... 40582 files and directories currently installed.)
Preparing to unpack .../0-liblmdb0_0.9.31-1build2_amd64.deb ...
Unpacking liblmdb0:amd64 (0.9.31-1build2) ...
Selecting previously unselected package liburcu8t64:amd64.
Preparing to unpack .../1-liburcu8t64_0.15.6-1_amd64.deb ...
Unpacking liburcu8t64:amd64 (0.15.6-1) ...
Selecting previously unselected package libuv1t64:amd64.
Preparing to unpack .../2-libuv1t64_1.51.0-2ubuntu1_amd64.deb ...
Unpacking libuv1t64:amd64 (1.51.0-2ubuntu1) ...
Selecting previously unselected package bind9-libs:amd64.
Preparing to unpack .../3-bind9-libs_1%3a9.20.24-1ubuntu0.5_amd64.deb ...
Unpacking bind9-libs:amd64 (1:9.20.24-1ubuntu0.5) ...
Selecting previously unselected package bind9-host.
Preparing to unpack .../4-bind9-host_1%3a9.20.24-1ubuntu0.5_amd64.deb ...
Unpacking bind9-host (1:9.20.24-1ubuntu0.5) ...
Selecting previously unselected package bind9-dnsutils.
Preparing to unpack .../5-bind9-dnsutils_1%3a9.20.24-1ubuntu0.5_amd64.deb ...
Unpacking bind9-dnsutils (1:9.20.24-1ubuntu0.5) ...
Preparing to unpack .../6-curl_8.18.0-1ubuntu2.8_amd64.deb ...
Unpacking curl (8.18.0-1ubuntu2.8) over (8.18.0-1ubuntu2.4) ...
Preparing to unpack .../7-libcurl4t64_8.18.0-1ubuntu2.8_amd64.deb ...
Unpacking libcurl4t64:amd64 (8.18.0-1ubuntu2.8) over (8.18.0-1ubuntu2.4) ...
Preparing to unpack .../8-libcurl3t64-gnutls_8.18.0-1ubuntu2.8_amd64.deb ...
Unpacking libcurl3t64-gnutls:amd64 (8.18.0-1ubuntu2.8) over (8.18.0-1ubuntu2.4) ...
Setting up liblmdb0:amd64 (0.9.31-1build2) ...
Setting up liburcu8t64:amd64 (0.15.6-1) ...
Setting up libcurl4t64:amd64 (8.18.0-1ubuntu2.8) ...
Setting up libuv1t64:amd64 (1.51.0-2ubuntu1) ...
Setting up libcurl3t64-gnutls:amd64 (8.18.0-1ubuntu2.8) ...
Setting up curl (8.18.0-1ubuntu2.8) ...
Setting up bind9-libs:amd64 (1:9.20.24-1ubuntu0.5) ...
Setting up bind9-host (1:9.20.24-1ubuntu0.5) ...
Setting up bind9-dnsutils (1:9.20.24-1ubuntu0.5) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
tejaswin@DESKTOP-P4G181S:~$
```

```
tejaswin@DESKTOP-P4G181S:~$ ip -br link
sudo tshark -D
lo               UNKNOWN        00:00:00:00:00:00 <LOOPBACK,UP,LOWER_UP>
eth0             DOWN           82:13:16:a1:19:fe <BROADCAST,MULTICAST>
loopback0        UP             00:15:5d:e3:85:90 <BROADCAST,MULTICAST,UP,LOWER_UP>
eth1             DOWN           80:13:16:a1:19:fe <BROADCAST,MULTICAST>
eth2             DOWN           80:13:16:a1:19:ff <BROADCAST,MULTICAST>
eth3             UP             d0:37:45:2e:63:10 <BROADCAST,MULTICAST,UP,LOWER_UP>
Running as user "root" and group "root". This could be dangerous.
1. loopback0
2. eth3
3. any
4. lo (Loopback)
5. bluetooth-monitor
6. nflog
7. nfqueue
8. dbus-system
9. dbus-session
10. eth0
11. eth1
12. eth2
13. ciscodump (Cisco remote capture)
14. dpauxmon (DisplayPort AUX channel monitor capture)
15. randpkt (Random packet generator)
16. sdjournal (systemd Journal Export)
17. sshdig (SSH remote syscall capture)
18. sshdump (SSH remote capture)
19. udpdump (UDP Listener remote capture)
20. wifidump (Wi-Fi remote capture)
```

```
tejaswin@DESKTOP-P4G181S:~$ sudo tshark -i eth0 -w ~/sender_capture.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
tshark: The capture session could not be initiated due to error getting information on pipe or socket: Permission denied.
```




```
tejaswin@DESKTOP-P4G181S:~$ sudo chmod 644 ~/sender_capture.pcapng
ls -lh ~/sender_capture.pcapng
chmod: cannot access '/home/tejaswin/sender_capture.pcapng': No such file or directory
ls: cannot access '/home/tejaswin/sender_capture.pcapng': No such file or directory
```

```
tejaswin@DESKTOP-P4G181S:~$ # 1. Display TCP packets
tshark -r ~/sender_capture.pcapng -Y "tcp" | head -n 15

# 2. Display UDP packets
tshark -r ~/sender_capture.pcapng -Y "udp" | head -n 15

# 3. Display ICMP packets
tshark -r ~/sender_capture.pcapng -Y "icmp"

# 4. Display TCP packets carrying application data
tshark -r ~/sender_capture.pcapng -Y "tcp.len > 0" | head -n 15
   13 1.333565190 10.250.24.52 → 142.251.157.119 TCP 74 45876 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3984199725 TSecr=0 WS=1024
   14 1.358259673 142.251.157.119 → 10.250.24.52 TCP 74 443 → 45876 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM TSval=499283310 TSecr=3984199725 WS=256
   15 1.358342699 10.250.24.52 → 142.251.157.119 TCP 66 45876 → 443 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=3984199750 TSecr=499283310
   16 1.360779359 10.250.24.52 → 142.251.157.119 TLSv1.2 1640 Client Hello (SNI=www.google.com)
   17 1.414862628 10.250.24.52 → 142.251.157.119 TCP 240 [TCP Retransmission] 45876 → 443 [PSH, ACK] Seq=1401 Ack=1 Win=64512 Len=174 TSval=3984199807 TSecr=499283310
   18 1.440433510 142.251.157.119 → 10.250.24.52 TLSv1.3 1466 Server Hello, Change Cipher Spec
   19 1.440434061 142.251.157.119 → 10.250.24.52 TCP 1466 443 → 45876 [PSH, ACK] Seq=1401 Ack=1575 Win=267520 Len=1400 TSval=499283344 TSecr=3984199753
   20 1.440434095 142.251.157.119 → 10.250.24.52 TLSv1.3 1141 Application Data
   21 1.440434126 142.251.157.119 → 10.250.24.52 TCP 78 [TCP Dup ACK 18#1] 443 → 45876 [ACK] Seq=3876 Ack=1575 Win=267520 Len=0 TSval=499283392 TSecr=3984199807 SLE=1401 SRE=1575
   22 1.440520235 10.250.24.52 → 142.251.157.119 TCP 66 45876 → 443 [ACK] Seq=1575 Ack=1401 Win=67584 Len=0 TSval=3984199832 TSecr=499283310
   23 1.440528669 10.250.24.52 → 142.251.157.119 TCP 66 45876 → 443 [ACK] Seq=1575 Ack=2801 Win=70656 Len=0 TSval=3984199832 TSecr=499283344
   24 1.440531183 10.250.24.52 → 142.251.157.119 TCP 66 45876 → 443 [ACK] Seq=1575 Ack=3876 Win=72704 Len=0 TSval=3984199832 TSecr=499283344
   25 1.443788742 142.251.157.119 → 10.250.24.52 TCP 1141 [TCP Spurious Retransmission] 443 → 45876 [PSH, ACK] Seq=2801 Ack=1575 Win=267520 Len=1075 TSval=499283395 TSecr=3984199807 [TCP PDU reassembled in 20]
   26 1.443821829 10.250.24.52 → 142.251.157.119 TCP 78 [TCP Dup ACK 24#1] 45876 → 443 [ACK] Seq=1575 Ack=3876 Win=72704 Len=0 TSval=3984199836 TSecr=499283395 SLE=2801 SRE=3876
   27 1.485793747 10.250.24.52 → 142.251.157.119 TLSv1.3 146 Change Cipher Spec, Application Data
   64 2.198236252 10.250.24.52 → 8.8.8.8      ICMP 98 Echo (ping) request  id=0x1f9c, seq=1/256, ttl=64
   68 2.256688897      8.8.8.8 → 10.250.24.52 ICMP 98 Echo (ping) reply    id=0x1f9c, seq=1/256, ttl=113 (request in 64)
   79 3.199874446 10.250.24.52 → 8.8.8.8      ICMP 98 Echo (ping) request  id=0x1f9c, seq=2/512, ttl=64
   80 3.280487546      8.8.8.8 → 10.250.24.52 ICMP 98 Echo (ping) reply    id=0x1f9c, seq=2/512, ttl=113 (request in 79)
   93 4.201676759 10.250.24.52 → 8.8.8.8      ICMP 98 Echo (ping) request  id=0x1f9c, seq=3/768, ttl=64
   94 4.220087619      8.8.8.8 → 10.250.24.52 ICMP 98 Echo (ping) reply    id=0x1f9c, seq=3/768, ttl=113 (request in 93)
  134 5.203501990 10.250.24.52 → 8.8.8.8      ICMP 98 Echo (ping) request  id=0x1f9c, seq=4/1024, ttl=64
  136 5.230120726      8.8.8.8 → 10.250.24.52 ICMP 98 Echo (ping) reply    id=0x1f9c, seq=4/1024, ttl=113 (request in 134)
   16 1.360779359 10.250.24.52 → 142.251.157.119 TLSv1.2 1640 Client Hello (SNI=www.google.com)
   17 1.414862628 10.250.24.52 → 142.251.157.119 TCP 240 [TCP Retransmission] 45876 → 443 [PSH, ACK] Seq=1401 Ack=1 Win=64512 Len=174 TSval=3984199807 TSecr=499283310
   18 1.440433510 142.251.157.119 → 10.250.24.52 TLSv1.3 1466 Server Hello, Change Cipher Spec
   19 1.440434061 142.251.157.119 → 10.250.24.52 TCP 1466 443 → 45876 [PSH, ACK] Seq=1401 Ack=1575 Win=267520 Len=1400 TSval=499283344 TSecr=3984199753
   20 1.440434095 142.251.157.119 → 10.250.24.52 TLSv1.3 1141 Application Data
   25 1.443788742 142.251.157.119 → 10.250.24.52 TCP 1141 [TCP Spurious Retransmission] 443 → 45876 [PSH, ACK] Seq=2801 Ack=1575 Win=267520 Len=1075 TSval=499283395 TSecr=3984199807 [TCP PDU reassembled in 20]
   27 1.485793747 10.250.24.52 → 142.251.157.119 TLSv1.3 146 Change Cipher Spec, Application Data
   28 1.485894491 10.250.24.52 → 142.251.157.119 TLSv1.3 197 Application Data
   29 1.486324168 10.250.24.52 → 142.251.157.119 TLSv1.3 114 Application Data
   30 1.555031860 10.250.24.52 → 142.251.157.119 TCP 114 [TCP Retransmission] 45876 → 443 [PSH, ACK] Seq=1786 Ack=3876 Win=72704 Len=48 TSval=3984199947 TSecr=499283395
   34 1.560060075 142.251.157.119 → 10.250.24.52 TLSv1.3 716 Application Data, Application Data
   35 1.560060106 142.251.157.119 → 10.250.24.52 TLSv1.3 97 Application Data
   38 1.560642916 10.250.24.52 → 142.251.157.119 TLSv1.3 97 Application Data
   46 1.952260038 142.251.157.119 → 10.250.24.52 TLSv1.3 1161 Application Data
   47 1.952606071 10.250.24.52 → 142.251.157.119 TLSv1.3 105 Application Data
```

```
tejaswin@DESKTOP-P4G181S:~$ # TCP Packet (Frame Len | Src MAC | Dst MAC | IP Hdr Len | IP Total Len | TCP Hdr Len | TCP Payload Len)
tshark -r ~/sender_capture.pcapng -Y "tcp.len > 0" -c 1 -V | head -n 45
tshark -r ~/sender_capture.pcapng -Y "tcp.len > 0" -c 1 -T fields \
  -e frame.len -e eth.src -e eth.dst -e ip.hdr_len -e ip.len -e tcp.hdr_len -e tcp.len

# UDP Packet (Frame Len | Src MAC | Dst MAC | IP Hdr Len | IP Total Len | UDP Total Len)
tshark -r ~/sender_capture.pcapng -Y "udp" -c 1 -V | head -n 40
tshark -r ~/sender_capture.pcapng -Y "udp" -c 1 -T fields \
  -e frame.len -e eth.src -e eth.dst -e ip.hdr_len -e ip.len -e udp.length
```

CNNP Skill 2

Name: Tejaswin Amara

Roll No.: 2520090104

```
Ubuntu
tejaswin@DESKTOP-P4G181S:~$ sudo apt update
sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [717 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [16.0 MB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [157 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [220 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [72.5 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [62.7 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3552 B]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [586 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [122 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [20.0 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [3728 B]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Components [212 B]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 c-n-f Metadata [120 B]
Get:19 http://archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Components [4556 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute/universe amd64 c-n-f Metadata [313 kB]
Get:22 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Packages [290 kB]
Get:23 http://archive.ubuntu.com/ubuntu resolute/multiverse Translation-en [127 kB]
Get:24 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Components [50.0 kB]
Get:25 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 c-n-f Metadata [8276 B]
Get:26 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [868 kB]
Get:27 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [195 kB]
Get:28 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [110 kB]
Get:29 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [389 kB]
Get:30 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [122 kB]
Get:31 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Get:32 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 c-n-f Metadata [4704 B]
Get:33 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [602 kB]
Get:34 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [125 kB]
Get:35 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [21.0 kB]
Get:36 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [4168 B]
Get:37 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Components [216 B]
```

```
tejaswin@DESKTOP-P4G181S:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
sudo ip netns list
host2
router
host1
```

```
tejaswin@DESKTOP-P4G181S:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2

sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
```

```
tejaswin@DESKTOP-P4G181S:~$ # Configure host1
sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up

# Configure router
sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up

# Configure host2
sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
```

```
tejaswin@DESKTOP-P4G181S:~$ sudo ip netns exec router sysctl -w net.ipv4.ip
_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
```

```
tejaswin@DESKTOP-P4G181S:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.050 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.043 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.052 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2051ms
rtt min/avg/max/mdev = 0.043/0.048/0.052/0.003 ms
```

```
tejaswin@DESKTOP-P4G181S:~$ sudo ip netns exec router tc qdisc add dev
veth-r2 root netem delay 20ms loss 1%
sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 9 limit 1000 delay 20ms loss 1% seed 1619
7084392192552354
```

```
tejaswin@DESKTOP-P4G181S:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
13
```

```
tejaswin@DESKTOP-P4G181S:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
13
```

```
tejaswin@DESKTOP-P4G181S:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
14
```

```
tejaswin@DESKTOP-P4G181S:~$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
```

```
tejaswin@DESKTOP-P4G181S:~$ sudo ip netns exec host1 nc 10.0.2.2 1234
[sudo: authenticate] Password:
sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
```

```
tejaswin@DESKTOP-P4G181S:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
[sudo: authenticate] Password:
-rw-r--r-- 1 root root 1.7K Nov 12 08:03 /tmp/cap1.pcap
-rw-r--r-- 1 root root 1.7K Nov 12 08:03 /tmp/cap2.pcap
-rw-r--r-- 1 root root 1.7K Nov 12 08:04 /tmp/cap3.pcap
```

```
tejaswin@DESKTOP-P4G181S:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      3 63
      4 64
      3 63
      4 64
      4 63
      3 64
```

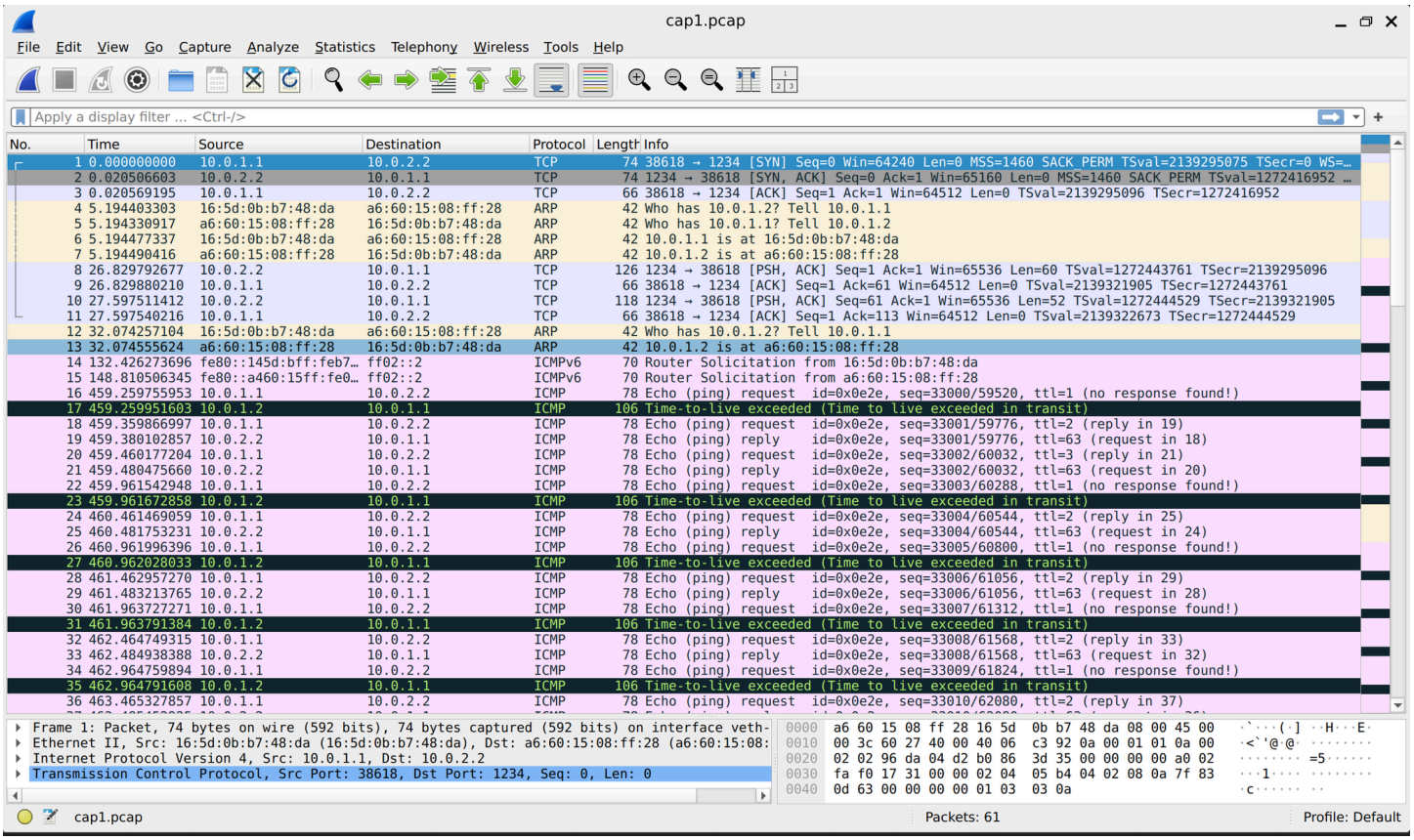
```
tejaswin@DESKTOP-P4G181S:~$ # View decoded ASCII text
tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e data.text 2>/dev/null

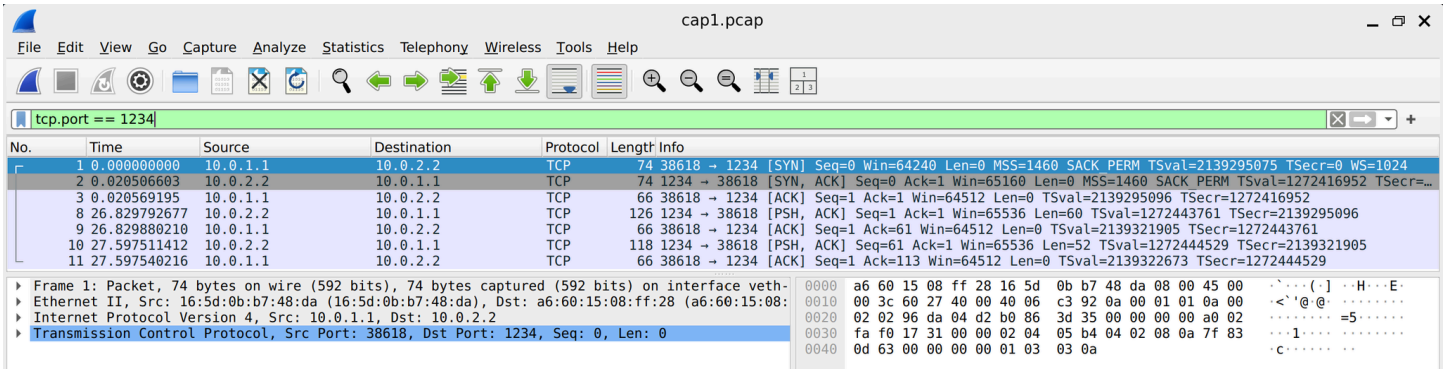
# View hex and ASCII dump
tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e tcp.payload 2>/dev/null | xxd | head -20
```

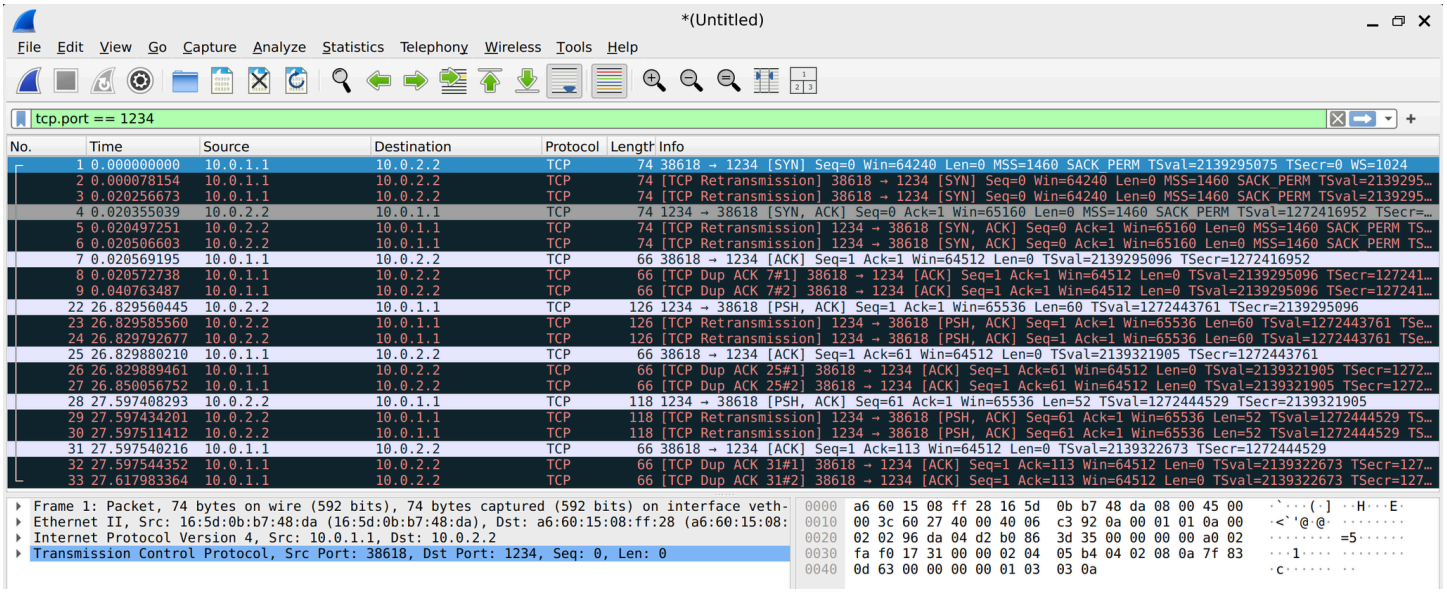
```
00000000: 0a0a 0a37 3337 3536 3436 6632 3036 3336  ...7375646f20636
00000010: 3836 6436 6636 3432 3033 3633 3433 3432  86d6f64203634342
00000020: 3032 6637 3436 6437 3032 6636 3336 3137  02f746d702f63617
00000030: 3033 3132 6537 3036 3336 3137 3032 3032  0312e70636170202
00000040: 6637 3436 6437 3032 6636 3336 3137 3033  f746d702f6361703
00000050: 3232 6537 3036 3336 3137 3032 3032 6637  22e70636170202f7
00000060: 3436 6437 3032 6636 3336 3137 3033 3332  46d702f636170332
00000070: 6537 3036 3336 3137 3030 610a 0a36 6337  e706361700a..6c7
00000080: 3332 3032 6436 6336 3832 3032 6637 3436  3202d6c68202f746
00000090: 6437 3032 6636 3336 3137 3033 3132 6537  d702f636170312e7
000000a0: 3036 3336 3137 3032 3032 6637 3436 6437  0636170202f746d7
000000b0: 3032 6636 3336 3137 3033 3232 6537 3036  02f636170322e706
000000c0: 3336 3137 3032 3032 6637 3436 6437 3032  36170202f746d702
000000d0: 6636 3336 3137 3033 3332 6537 3036 3336  f636170332e70636
000000e0: 3137 3030 610a 0a                        1700a..
```

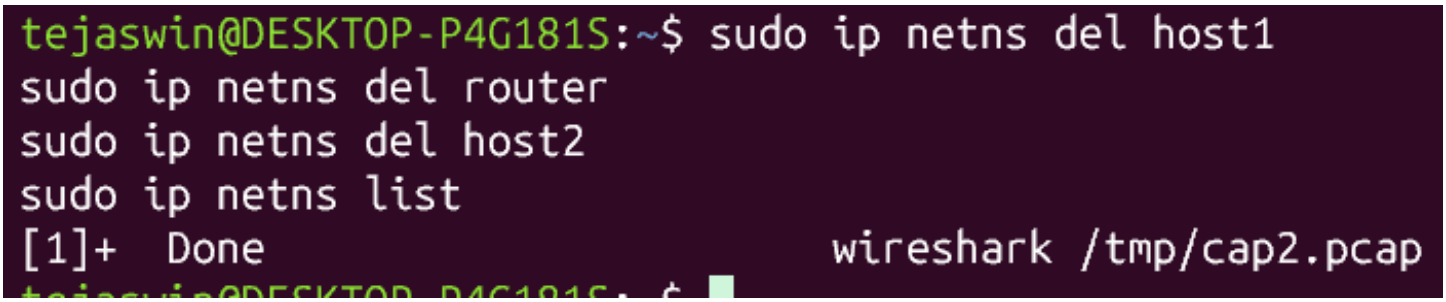
```
tejaswin@DESKTOP-P4G181S:~$ tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null
tshark -r /tmp/cap2.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null
tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null
0.000000000
0.000000000
0.000000000
```

```
tejaswin@DESKTOP-P4G181S:~$ sudo ip netns exec host1 mtr --report 10.0.2.2
Start: 2026-11-12T08:09:00+0000
HOST: DESKTOP-P4G181S             Loss%   Snt   Last   Avg  Best  Wrst StDev
  1.|-- 10.0.1.2                   0.0%    10    0.1   0.1   0.1   0.5   0.1
  2.|-- 10.0.2.2                   0.0%    10   20.4  20.3  20.2  20.4   0.1
tejaswin@DESKTOP-P4G181S:~$ wireshark /tmp/cap2.pcap &
[1] 3646
```


cap1.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000000 10.0.1.1 10.0.2.2 TCP 74 38618 → 1234 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2139295075 TSecr=0 WS=…
2 0.020506603 10.0.2.2 10.0.1.1 TCP 74 1234 → 38618 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=1272416952 …
3 0.020569195 10.0.1.1 10.0.2.2 TCP 66 38618 → 1234 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=2139295096 TSecr=1272416952
4 5.194403303 16:5d:0b:b7:48:da a6:60:15:08:ff:28 ARP 42 Who has 10.0.1.2? Tell 10.0.1.1
5 5.194330917 a6:60:15:08:ff:28 16:5d:0b:b7:48:da ARP 42 Who has 10.0.1.1? Tell 10.0.1.2
6 5.194477337 16:5d:0b:b7:48:da a6:60:15:08:ff:28 ARP 42 10.0.1.1 is at 16:5d:0b:b7:48:da
7 5.194490416 a6:60:15:08:ff:28 16:5d:0b:b7:48:da ARP 42 10.0.1.2 is at a6:60:15:08:ff:28
8 26.829792677 10.0.2.2 10.0.1.1 TCP 126 1234 → 38618 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=60 TSval=1272443761 TSecr=2139295096
9 26.829880210 10.0.1.1 10.0.2.2 TCP 66 38618 → 1234 [ACK] Seq=1 Ack=61 Win=64512 Len=0 TSval=2139321905 TSecr=1272443761
10 27.597511412 10.0.2.2 10.0.1.1 TCP 118 1234 → 38618 [PSH, ACK] Seq=61 Ack=1 Win=65536 Len=52 TSval=1272444529 TSecr=2139321905
11 27.597540216 10.0.1.1 10.0.2.2 TCP 66 38618 → 1234 [ACK] Seq=1 Ack=113 Win=64512 Len=0 TSval=2139322673 TSecr=1272444529
12 32.074257104 16:5d:0b:b7:48:da a6:60:15:08:ff:28 ARP 42 Who has 10.0.1.2? Tell 10.0.1.1
13 32.074555624 a6:60:15:08:ff:28 16:5d:0b:b7:48:da ARP 42 10.0.1.2 is at a6:60:15:08:ff:28
14 132.426273696 fe80::145d:bff:feb7… ff02::2 ICMPv6 70 Router Solicitation from 16:5d:0b:b7:48:da
15 148.810506345 fe80::a460:15ff:fe0… ff02::2 ICMPv6 70 Router Solicitation from a6:60:15:08:ff:28
16 459.259755953 10.0.1.1 10.0.2.2 ICMP 78 Echo (ping) request id=0x0e2e, seq=33000/59520, ttl=1 (no response found!)
17 459.259951603 10.0.1.2 10.0.1.1 ICMP 106 Time-to-live exceeded (Time to live exceeded in transit)
18 459.359866997 10.0.1.1 10.0.2.2 ICMP 78 Echo (ping) request id=0x0e2e, seq=33001/59776, ttl=2 (reply in 19)
19 459.380102857 10.0.2.2 10.0.1.1 ICMP 78 Echo (ping) reply id=0x0e2e, seq=33001/59776, ttl=63 (request in 18)
20 459.460177204 10.0.1.1 10.0.2.2 ICMP 78 Echo (ping) request id=0x0e2e, seq=33002/60032, ttl=3 (reply in 21)
21 459.480475660 10.0.2.2 10.0.1.1 ICMP 78 Echo (ping) reply id=0x0e2e, seq=33002/60032, ttl=63 (request in 20)
22 459.961542948 10.0.1.1 10.0.2.2 ICMP 78 Echo (ping) request id=0x0e2e, seq=33003/60288, ttl=1 (no response found!)
23 459.961672858 10.0.1.2 10.0.1.1 ICMP 106 Time-to-live exceeded (Time to live exceeded in transit)
24 460.461469059 10.0.1.1 10.0.2.2 ICMP 78 Echo (ping) request id=0x0e2e, seq=33004/60544, ttl=2 (reply in 25)
25 460.481753231 10.0.2.2 10.0.1.1 ICMP 78 Echo (ping) reply id=0x0e2e, seq=33004/60544, ttl=63 (request in 24)
26 460.961996396 10.0.1.1 10.0.2.2 ICMP 78 Echo (ping) request id=0x0e2e, seq=33005/60800, ttl=1 (no response found!)
27 460.962028033 10.0.1.2 10.0.1.1 ICMP 106 Time-to-live exceeded (Time to live exceeded in transit)
28 461.462957270 10.0.1.1 10.0.2.2 ICMP 78 Echo (ping) request id=0x0e2e, seq=33006/61056, ttl=2 (reply in 29)
29 461.483213765 10.0.2.2 10.0.1.1 ICMP 78 Echo (ping) reply id=0x0e2e, seq=33006/61056, ttl=63 (request in 28)
30 461.963727271 10.0.1.1 10.0.2.2 ICMP 78 Echo (ping) request id=0x0e2e, seq=33007/61312, ttl=1 (no response found!)
31 461.963791384 10.0.1.2 10.0.1.1 ICMP 106 Time-to-live exceeded (Time to live exceeded in transit)
32 462.464749315 10.0.1.1 10.0.2.2 ICMP 78 Echo (ping) request id=0x0e2e, seq=33008/61568, ttl=2 (reply in 33)
33 462.484938388 10.0.2.2 10.0.1.1 ICMP 78 Echo (ping) reply id=0x0e2e, seq=33008/61568, ttl=63 (request in 32)
34 462.964759894 10.0.1.1 10.0.2.2 ICMP 78 Echo (ping) request id=0x0e2e, seq=33009/61824, ttl=1 (no response found!)
35 462.964791608 10.0.1.2 10.0.1.1 ICMP 106 Time-to-live exceeded (Time to live exceeded in transit)
36 463.465327857 10.0.1.1 10.0.2.2 ICMP 78 Echo (ping) request id=0x0e2e, seq=33010/62080, ttl=2 (reply in 37)
Frame 1: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface veth-
Ethernet II, Src: 16:5d:0b:b7:48:da (16:5d:0b:b7:48:da), Dst: a6:60:15:08:ff:28 (a6:60:15:08:
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 10.0.2.2
Transmission Control Protocol, Src Port: 38618, Dst Port: 1234, Seq: 0, Len: 0
0000 a6 60 15 08 ff 28 16 5d 0b b7 48 da 08 00 45 00
0010 00 3c 60 27 40 00 40 06 c3 92 0a 00 01 01 0a 00
0020 02 02 96 da 04 d2 b0 86 3d 35 00 00 00 00 a0 02
0030 fa f0 17 31 00 00 02 04 05 b4 04 02 08 0a 7f 83
0040 0d 63 00 00 00 00 01 03 03 0a
cap1.pcap
Packets: 61
Profile: Default


cap1.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.port == 1234
No. Time Source Destination Protocol Length Info
1 0.000000000 10.0.1.1 10.0.2.2 TCP 74 38618 → 1234 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2139295075 TSecr=0 WS=1024
2 0.020506603 10.0.2.2 10.0.1.1 TCP 74 1234 → 38618 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=1272416952 TSecr=…
3 0.020569195 10.0.1.1 10.0.2.2 TCP 66 38618 → 1234 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=2139295096 TSecr=1272416952
8 26.829792677 10.0.2.2 10.0.1.1 TCP 126 1234 → 38618 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=60 TSval=1272443761 TSecr=2139295096
9 26.829880210 10.0.1.1 10.0.2.2 TCP 66 38618 → 1234 [ACK] Seq=1 Ack=61 Win=64512 Len=0 TSval=2139321905 TSecr=1272443761
10 27.597511412 10.0.2.2 10.0.1.1 TCP 118 1234 → 38618 [PSH, ACK] Seq=61 Ack=1 Win=65536 Len=52 TSval=1272444529 TSecr=2139321905
11 27.597540216 10.0.1.1 10.0.2.2 TCP 66 38618 → 1234 [ACK] Seq=1 Ack=113 Win=64512 Len=0 TSval=2139322673 TSecr=1272444529
Frame 1: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface veth-
Ethernet II, Src: 16:5d:0b:b7:48:da (16:5d:0b:b7:48:da), Dst: a6:60:15:08:ff:28 (a6:60:15:08:
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 10.0.2.2
Transmission Control Protocol, Src Port: 38618, Dst Port: 1234, Seq: 0, Len: 0
0000 a6 60 15 08 ff 28 16 5d 0b b7 48 da 08 00 45 00
0010 00 3c 60 27 40 00 40 06 c3 92 0a 00 01 01 0a 00
0020 02 02 96 da 04 d2 b0 86 3d 35 00 00 00 00 a0 02
0030 fa f0 17 31 00 00 02 04 05 b4 04 02 08 0a 7f 83
0040 0d 63 00 00 00 00 01 03 03 0a


*(Untitled)
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.port == 1234
No. Time Source Destination Protocol Length Info
1 0.000000000 10.0.1.1 10.0.2.2 TCP 74 38618 → 1234 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2139295075 TSecr=0 WS=1024
2 0.000078154 10.0.1.1 10.0.2.2 TCP 74 [TCP Retransmission] 38618 → 1234 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2139295…
3 0.020256673 10.0.1.1 10.0.2.2 TCP 74 [TCP Retransmission] 38618 → 1234 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2139295…
4 0.020355039 10.0.2.2 10.0.1.1 TCP 74 1234 → 38618 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=1272416952 TSecr=…
5 0.020497251 10.0.2.2 10.0.1.1 TCP 74 [TCP Retransmission] 1234 → 38618 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TS…
6 0.020506603 10.0.2.2 10.0.1.1 TCP 74 [TCP Retransmission] 1234 → 38618 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TS…
7 0.020569195 10.0.1.1 10.0.2.2 TCP 66 38618 → 1234 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=2139295096 TSecr=1272416952
8 0.020572738 10.0.1.1 10.0.2.2 TCP 66 [TCP Dup ACK 7#1] 38618 → 1234 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=2139295096 TSecr=127241…
9 0.040763487 10.0.1.1 10.0.2.2 TCP 66 [TCP Dup ACK 7#2] 38618 → 1234 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=2139295096 TSecr=127241…
22 26.829560445 10.0.2.2 10.0.1.1 TCP 126 1234 → 38618 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=60 TSval=1272443761 TSecr=2139295096
23 26.829585560 10.0.2.2 10.0.1.1 TCP 126 [TCP Retransmission] 1234 → 38618 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=60 TSval=1272443761 TSe…
24 26.829792677 10.0.2.2 10.0.1.1 TCP 126 [TCP Retransmission] 1234 → 38618 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=60 TSval=1272443761 TSe…
25 26.829880210 10.0.1.1 10.0.2.2 TCP 66 38618 → 1234 [ACK] Seq=1 Ack=61 Win=64512 Len=0 TSval=2139321905 TSecr=1272443761
26 26.829889461 10.0.1.1 10.0.2.2 TCP 66 [TCP Dup ACK 25#1] 38618 → 1234 [ACK] Seq=1 Ack=61 Win=64512 Len=0 TSval=2139321905 TSecr=1272…
27 26.850056752 10.0.1.1 10.0.2.2 TCP 66 [TCP Dup ACK 25#2] 38618 → 1234 [ACK] Seq=1 Ack=61 Win=64512 Len=0 TSval=2139321905 TSecr=1272…
28 27.597408293 10.0.2.2 10.0.1.1 TCP 118 1234 → 38618 [PSH, ACK] Seq=61 Ack=1 Win=65536 Len=52 TSval=1272444529 TSecr=2139321905
29 27.597434201 10.0.2.2 10.0.1.1 TCP 118 [TCP Retransmission] 1234 → 38618 [PSH, ACK] Seq=61 Ack=1 Win=65536 Len=52 TSval=1272444529 TS…
30 27.597511412 10.0.2.2 10.0.1.1 TCP 118 [TCP Retransmission] 1234 → 38618 [PSH, ACK] Seq=61 Ack=1 Win=65536 Len=52 TSval=1272444529 TS…
31 27.597540216 10.0.1.1 10.0.2.2 TCP 66 38618 → 1234 [ACK] Seq=1 Ack=113 Win=64512 Len=0 TSval=2139322673 TSecr=1272444529
32 27.597544352 10.0.1.1 10.0.2.2 TCP 66 [TCP Dup ACK 31#1] 38618 → 1234 [ACK] Seq=1 Ack=113 Win=64512 Len=0 TSval=2139322673 TSecr=127…
33 27.617983364 10.0.1.1 10.0.2.2 TCP 66 [TCP Dup ACK 31#2] 38618 → 1234 [ACK] Seq=1 Ack=113 Win=64512 Len=0 TSval=2139322673 TSecr=127…
Frame 1: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface veth-
Ethernet II, Src: 16:5d:0b:b7:48:da (16:5d:0b:b7:48:da), Dst: a6:60:15:08:ff:28 (a6:60:15:08:
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 10.0.2.2
Transmission Control Protocol, Src Port: 38618, Dst Port: 1234, Seq: 0, Len: 0
0000 a6 60 15 08 ff 28 16 5d 0b b7 48 da 08 00 45 00
0010 00 3c 60 27 40 00 40 06 c3 92 0a 00 01 01 0a 00
0020 02 02 96 da 04 d2 b0 86 3d 35 00 00 00 00 a0 02
0030 fa f0 17 31 00 00 02 04 05 b4 04 02 08 0a 7f 83
0040 0d 63 00 00 00 00 01 03 03 0a


tejaswin@DESKTOP-P4G181S:~$ sudo ip netns del host1
sudo ip netns del router
sudo ip netns del host2
sudo ip netns list
[1]+ Done wireshark /tmp/cap2.pcap

CNNP Practice Questions

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